

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.3: Matrix-vector product and linear maps

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Vectors as columns

A vector can be written as a row or as a column. When a matrix acts on a vector in this course, we usually write the vector as a column:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

From now on, when we write $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x}$ is interpreted as a column vector.

If A is an $m \times n$ matrix, then A has n columns. We write

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$$

where each $\mathbf{a}_j$ is a column vector in $\mathbb{R}^m$.

Matrix-vector product

Definition: Matrix-vector product Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ be an $m \times n$ matrix and let $\mathbf{x} = (x_1, \dots, x_n)$ be a column vector in $\mathbb{R}^n$. The

matrix-vector product is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Activity: Computing a matrix-vector product

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Using columns,

$$\begin{aligned} A\mathbf{x} &= 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} 4 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} -5 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -34 \\ -5 \end{bmatrix}. \end{aligned}$$

Theorem: Properties of matrix-vector products Let A and B be $m \times n$ matrices, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let r be a scalar. Then

- $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$,
- $A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}$,
- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Theorem: Row dot-product view If A is an $m \times n$ matrix with rows $\text{row}_1(A), \dots, \text{row}_m(A)$, then

$$A\mathbf{x} = \begin{bmatrix} \text{row}_1(A) \cdot \mathbf{x} \\ \text{row}_2(A) \cdot \mathbf{x} \\ \vdots \\ \text{row}_m(A) \cdot \mathbf{x} \end{bmatrix}.$$

Note: Rows measure; columns contribute A matrix-vector product has two complementary readings. In the row view, each row measures the input by taking a dot product with $\mathbf{x}$. In the column view,

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

so the input coordinates tell how much each column contributes to the output.

Examples of matrix-vector products

The same product $A\mathbf{x}$ can represent different operations, depending on what the rows and columns of A mean. The next activities show three common patterns: scoring, selecting, and measuring changes.

Activity: Scoring objects by features

Let the rows of X represent three objects with two features:

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix}, \quad \mathbf{w} = (2, -1).$$

Compute $X\mathbf{w}$. Interpret the result.

We have

$$X\mathbf{w} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}.$$

Each output is a score:

$$2(\text{first feature}) - (\text{second feature}).$$

Activity: A selector matrix

Let

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

Compute $S\mathbf{x}$. What does S do?

We have

$$\begin{aligned} S\mathbf{x} &= \begin{bmatrix} 1 \cdot 5 + 0 \cdot 6 + 0 \cdot 7 + 0 \cdot 8 \\ 0 \cdot 5 + 0 \cdot 6 + 1 \cdot 7 + 0 \cdot 8 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 7 \end{bmatrix}. \end{aligned}$$

The matrix S selects the first and third entries of $\mathbf{x}$.

Activity: A difference matrix

Let

$$D = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

Compute $D\mathbf{x}$. What does D measure?

We have

$$\begin{aligned} D\mathbf{x} &= \begin{bmatrix} -1 \cdot 2 + 1 \cdot 5 + 0 \cdot 9 \\ 0 \cdot 2 + (-1) \cdot 5 + 1 \cdot 9 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \end{aligned}$$

The entries are consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4.$$

Thus D measures changes in a short time series.

Activity: Document ranking in matrix code

In Word-count vectors, we ranked documents by cosine similarity one score at a time. The following code repeats that ranking using a data matrix whose rows are the document vectors.

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])

X = np.array([
    [2, 0, 2],
    [1, 1, 0],
    [0, 2, 0],
], dtype=float)

q = np.array([1, 0, 1], dtype=float)

scores = (X @ q) / (np.linalg.norm(X, axis=1) * np.linalg.norm(q))
ranking = doc_names[np.argsort(scores)[::-1]]

scores, ranking
```

Output:

```
(array([1. , 0.5, 0. ]),
 array(['D1', 'D2', 'D3'], dtype='<U2'))
```

1. Which row of X is the vector $\mathbf{D}_2$?
2. Which expression computes the three dot products $\mathbf{D}_i \cdot \mathbf{q}$?
3. What does `np.argsort(scores)[::-1]` do?
4. Why does the output agree with Word-count vectors?

The rows of X are D_1 , D_2 , and D_3 . The product $X @ q$ computes the three dot products with the query. The expression `np.argsort(scores)[::-1]` gives the indices of the scores from largest to smallest. The output agrees with Word-count vectors: $1, \frac{1}{2}, 0$.

Example: The attention activity in matrix form In A tiny attention-style token-weighting calculation, we computed the scores

$$q \cdot k_{\text{small}}, \quad q \cdot k_{\text{red}}, \quad q \cdot k_{\text{bird}}$$

one at a time. Matrix-vector multiplication computes the same scores at once.

Put the key vectors as the rows of

$$K = \begin{bmatrix} k_{\text{small}}^T \\ k_{\text{red}}^T \\ k_{\text{bird}}^T \end{bmatrix}.$$

Then

$$Kq = \begin{bmatrix} k_{\text{small}} \cdot q \\ k_{\text{red}} \cdot q \\ k_{\text{bird}} \cdot q \end{bmatrix}.$$

For the numbers in A tiny attention-style token-weighting calculation,

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$K\mathbf{q} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Activity: The same attention calculation in code

The following code performs the calculation from the token attention activity.

```
import numpy as np

tokens = np.array(["small", "red", "bird"])

q = np.array([1.0, 1.0])

K = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])

V = np.array([
    [4.0, 0.0],
    [0.0, 4.0],
    [4.0, 4.0],
])

s = K @ q
alpha = s / s.sum()
out = alpha @ V

s, alpha, out
```

Output:

```
(array([1., 1., 2.]),
 array([0.25, 0.25, 0.5 ]),
 array([3., 3.]))
```

1. Which line computes the token scores?
2. Which token receives the largest weight?
3. Which line forms the weighted¹⁰ average?

The line $s = K @ q$ computes the three dot products. The token “bird” receives the largest weight. The line $out = alpha @ V$ forms the weighted average of the rows of V . The entries of $alpha$ add to 1 because the scores were divided by their sum.

Geometric matrix actions in $\mathbb{R}^2$

Definition: Identity matrix The

identity matrix

I_n is the $n \times n$ matrix with 1s on the main diagonal and zeros elsewhere.

Fact For every column vector $x \in \mathbb{R}^n$, $I_n x = x$.

Definition: Standard basis The j -th

standard basis vector

e_j is the column vector with a 1 in position j and zeros elsewhere.

Fact If $A = [a_1 \ \cdots \ a_n]$, then $Ae_j = a_j$. Applying a matrix to a standard basis vector picks out a column.

In $\mathbb{R}^2$, a matrix $A = [a_1 \ a_2]$ sends e_1 to a_1 and e_2 to a_2 . Since every vector $x = (x_1, x_2)$ can be written as $x_1 e_1 + x_2 e_2$, we have

$$Ax = x_1 a_1 + x_2 a_2.$$

Thus a 2×2 matrix is geometrically determined by where it sends the two coordinate directions.

To understand a 2×2 matrix A , apply it to the four corners of the unit square. The image is usually a parallelogram. The two edges leaving the origin are the columns of A . We will refer to this as the

unit-square visualization

• **Matrix:** $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: horizontal stretch

Unit-square effect: The right edge moves farther right.

• **Matrix:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: projection onto the y -axis

Unit-square effect: Horizontal information is forgotten.

• **Matrix:** $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: reflection across the y -axis

Unit-square effect: The square flips left.

- **Matrix:** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Name: horizontal shear

Unit-square effect: The bottom edge stays fixed; the top edge shifts right.

- **Matrix:** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Name: 90° counterclockwise rotation

Unit-square effect: The square rotates around the origin.

- **Matrix:** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Name: projection onto the x -axis

Unit-square effect: Vertical information is forgotten.

For more illustrations of these transformations, see Lab U1: Vectors, Similarity, Attention, and Matrix Actions.

Example: Which column tells the story? For each matrix in the compact transformation gallery, compare Ae_1 and Ae_2 with e_1 and e_2 . Which column changed most visibly from the identity matrix? What does that change do to the unit square?

The columns $A\mathbf{e}_1$ and $A\mathbf{e}_2$ are the transformed coordinate directions. A stretch changes the length of a column. A reflection reverses the direction of a column. A shear changes one column by adding part of the other direction. A projection sends one column to zero, so one direction is forgotten.

Activity: Reflection across the y-axis

Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{x}$, and $A\mathbf{y}$, where $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (0, 1)$. Which coordinate changes?

We get

$$A\mathbf{e}_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad A\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The first coordinate flips sign and the second coordinate is unchanged.

Activity: Horizontal shear

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, and $A\mathbf{x}$, where $\mathbf{x} = (1, 1)$. Explain why the bottom edge stays fixed and the top edge shifts right.

We have

$$A\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The formula is $A(x, y) = (x + y, y)$. Points with $y = 0$ stay fixed, and points with $y = 1$ shift right by 1.

Note For the shear matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the row view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1, 1) \cdot (x, y) \\ (0, 1) \cdot (x, y) \end{bmatrix} = \begin{bmatrix} x + y \\ y \end{bmatrix}.$$

The first row measures horizontal plus vertical, while the second row measures vertical only. The column view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the y -coordinate contributes both upward and rightward motion.

Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition: Matrix map Given an $m \times n$ matrix A , the induced by A is the function $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by

matrix map

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition: Linear map A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is

linear

if

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v}), \\ T(c\mathbf{v}) &= cT(\mathbf{v}) \end{aligned}$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c .

Theorem: Matrix maps are linear If A is an $m \times n$ matrix, then the function $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ is linear.

Proof Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \quad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Theorem: Every linear map has a matrix Every linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Activity: Images of basis vectors determine the matrix

Suppose a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (-1, 3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

The columns are the images of the standard basis vectors, so

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

Warning We use “linear map” and “linear transformation” interchangeably; “transformation” is often used when emphasizing geometry.

Definition: Affine map A function of the form

$$\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$$

is called

affine

. It is built from a linear map followed by a translation.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

Example: An affine layer A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.